



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2024

Paper: Mathematical Methods of Physics-II

Course Code: PHYS-3503

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following:

(6x5=30)

1) The electric is defined as the gradient of a scalar potential ϕ , and has components $E_i = -\frac{\partial \phi}{\partial x_i}$.

Prove that $\vec{E}$ is a first order tensor.

2) Calculate the elements g_{ij} of the metric tensor for cylindrical polar coordinates.

3) Show that the complex numbers $1, i, -1, -i$ form a cyclic group of order 3.

4) Show that the set $S = \{1, 3, 5, 7\}$ is a group under multiplication (mod 8).

5) What is meant by analytic function? Show that $f(z) = z^2$ is analytic function.

6) By using Cauchy's integral theorem, Show that

$$\oint_C \frac{dz}{z^2 + z} = 0$$

in which the contour C is a circle defined by $|z| = R > 1$.

Q.2. Solve the following.

(5x6=30)

1) Show that the three matrices $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, C = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$, form an abelian group of order 3 under matrix multiplication.

2) Show that the quantities $g_{ij} = e_i \cdot e_j$ form the covariant components of a second-order tensor.

3) Show that the covariant derivative of a vector v is given by

$$v_{;j}^i = \frac{\partial v^i}{\partial u^j} + \Gamma_{kj}^i v^k$$

4) By using the Residue theorem, Evaluate the integral

$$\int_0^{2\pi} \frac{\cos 2\theta d\theta}{5 - 4\cos\theta}$$

5) Evaluate

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx$$

$$\frac{1}{2} \left(2 + \frac{1}{2} \right)$$

$$\frac{1}{2} \left(2 + \frac{1}{2} \right)$$

$$z = u + iv$$

$$z = u + iv$$

$$\frac{5}{4} \frac{3}{4}$$

$$\frac{2}{4} \frac{1}{2}$$

$$\frac{-2}{-4} \frac{1}{2}$$

$$\frac{2}{4} \frac{1}{2}$$

$$\frac{2}{4} \frac{1}{2}$$

MMP Spring 2024

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Short Q's:-

- 1 The electric is defined as the gradient of a scalar potential ϕ , and has components $E_i = -\partial\phi$. Prove that E is a first order tensor.

Solve:-

$$E'_i = \frac{\partial x_j}{\partial x'_i} E_j \quad (\text{we have to prove this})$$

So,

Since ϕ is a scalar. It is invariant under co-ordinate transformation.

$$\phi'(x') = \phi(x)$$

So

$$E'_i = -\frac{\partial\phi}{\partial x'_i}$$

$$E'_i = -\frac{\partial\phi}{\partial x_j} \cdot \frac{\partial x_j}{\partial x'_i}$$

$$E'_i = \frac{\partial x_j}{\partial x'_i} \left(-\frac{\partial\phi}{\partial x_j} \right)$$

$$E'_i = \frac{\partial x_j}{\partial x'_i} E_j \quad \text{Proved}$$

E_i is a first order tensor

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QNO 2:-

Calculate the element g_{ij} of the metric tensor for cylindrical polar co-ordinates.

The cylindrical polar co-ordinates are (ρ, ϕ, z) are related to cartesian co-ordinates (x, y, z) .

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

The differential line element ds^2 in cylindrical co-ordinates is

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2$$

In matrix form

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

coordinates are ordered as (ρ, ϕ, z)

→ $g_{\rho\rho} = 1$

→ $g_{\phi\phi} = \rho^2$

→ $g_{zz} = 1$

Q No 3:- Show that the complex number $1, i, -1, -i$ form a cyclic group of order 4.

$$\text{Let } C_4 = (1, -1, i, -i)$$

$\cdot$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

' $\cdot$ ' is associative by Table.

1 is its own inverse.

-1 is its own inverse.

i is inverse of -i

-i is inverse of i

So C_4 is group under ' $\cdot$ '

Also ' $\cdot$ ' is commutative so

C_4 is also abelian group.

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QNO 4:- Show that the set $= \{1, 3, 5, 7\}$ is a group under multiplication (mod 8).

Solve:-

•	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

- It is defined.
- It is multiplicative identity.

$$3 \cdot (5 \cdot 7) = (3 \cdot 5) \cdot 7$$

$$3 \cdot 3 = 7 \cdot 7$$

$$1 = 1$$

1 = is its own inverse

3 " " "

5 " "

7 " "

- Clearly hold associative property.
- Identity element 1 belong to G .
- Inverse of each element of G is also in G .

Hence G is an abelian group under multiplication.

$$\begin{array}{r} 35 \\ 38 \\ \hline 3 \end{array}$$

Qnos:-

What is meant by analytic function?
Show that $f(z) = z^2$ is analytic function.

"An analytic function (also called holomorphic) is a complex valued function that is differentiable at every point in some open region of the complex plane.

If a function is complex differentiable at every point in a neighbourhood of a point then it is analytic".
then

$$f(z) = z^2$$

$$z = x + iy$$

$$f(z) = (x + iy)^2 \\ = x^2 - y^2 + 2xyi$$

let

$$u = x^2 - y^2, \quad v = 2xy$$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2x$$

$$\text{Now: } \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial x} = 2y$$

$$\text{So } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\boxed{\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

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Hence proved It is analytical.

Qno 6:- By using Cauchy's integral theorem, Show that

$$\oint_C \frac{dz}{z^2 + z} = 0$$

in which the contour C is a circle defined by $|z| = R > 1$.

Soln:

$$\oint \frac{dz}{z^2 + z} = 0$$

$z=0$, $z=-1$ (poles)

$$|z| = R < 1$$

The poles at $z=0$, But does not contain $z=-1$

$$\frac{1}{z(z+1)} = \frac{A}{z} + \frac{B}{z+1} \quad (\text{By Partial fraction})$$

$$\frac{1}{z(z+1)} = \frac{1}{z} - \frac{1}{z+1}$$

$$\oint \frac{1}{z(z+1)} dz = \oint \left(\frac{1}{z} - \frac{1}{z+1} \right) dz$$

$$\oint \frac{1}{z(z+1)} dz = 2\pi i - 0$$

$$\boxed{\oint \frac{1}{z(z+1)} dz = 2\pi i}$$

Long Questions:-

1 Show that the three matrices
 $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$, $C = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$
 form an abelian group of order 3
 under matrix multiplication.

Solve:

$$B \cdot C = A$$

$$C \cdot B = A$$

$$B \cdot C = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$$

$$B \cdot C = \begin{pmatrix} 0 \times (-1) + 1 \times 1 & 0 \times (-1) + 1 \times 0 \\ -1 \times (-1) + (-1) \times 1 & -1 \times (-1) + (-1) \times 0 \end{pmatrix}$$

$$B \cdot C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A$$

Now

~~B.C~~

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$$C.B = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$$

$$C.B = \begin{pmatrix} -1 \times 0 + (-1) \times (-1) & -1 \times 1 + 1 \times -1 \\ 1 \times 0 + 0 \times (-1) & 1 \times 1 + 0 \times (-1) \end{pmatrix}$$

$$C.B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A$$

$$A = I$$

$$A.B = B.A = B.$$

$$A.C = C.A = C.$$

- A is its own inverse
- $B.C = A$, $C = B^{-1}$, $B = C^{-1}$

So all element have inverse in the set.

- Matrix multiplication always associative.
- Commutability holds, so the group is abelian.

Long Qno 2:-

Show that $g_{ij} = e_i \cdot e_j$ from the co-variant components of a second order tensor.

Soln:

$$g_{ij} = e_i \cdot e_j$$

where e_i and e_j are basis vector
let there be a coordinate transformation.

$$\bar{x}^k = \bar{x}^k(x^1, x^2, x^3)$$

basis vector transformation as

$$\bar{e}_k = \frac{\partial x^i}{\partial \bar{x}^k} e_i, \quad \bar{e}_l = \frac{\partial x^j}{\partial \bar{x}^l} e_j$$

Now,

$$\begin{aligned} \bar{g}_{kl} &= \bar{e}_k \cdot \bar{e}_l \\ &= \left(\frac{\partial x^i}{\partial \bar{x}^k} e_i \right) \cdot \left(\frac{\partial x^j}{\partial \bar{x}^l} e_j \right) \end{aligned}$$

$$\bar{g}_{kl} = \frac{\partial x^i}{\partial \bar{x}^k} \cdot \frac{\partial x^j}{\partial \bar{x}^l} (e_i \cdot e_j)$$

$$\bar{g}_{kl} = \frac{\partial x^i}{\partial \bar{x}^k} \cdot \frac{\partial x^j}{\partial \bar{x}^l} g_{ij}$$

This is co-variant components of a second order tensor.

QNO-3

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Show that covariant derivative of a vector v is given by

$$v_{ij}^i = \frac{\partial v^i}{\partial u^j} + \Gamma_{kj}^i v^k$$

Soln:

let: $V = v^i e_i$ ~~at point P on manifold~~

Diff b/s with r.to coordinate u^i .

$$\frac{\partial V}{\partial u^j} = \frac{\partial v^i}{\partial u^j} e_i + v^i \frac{\partial e_i}{\partial u^j} \quad \text{--- (A)}$$

$$\frac{\partial e_i}{\partial u^j} = \Gamma_{ij}^k e_k \quad \text{in Christoffel symbol}$$

Therefore:

$$v^i \frac{\partial e_i}{\partial u^j} = v^i \Gamma_{ij}^k e_k$$

change the dummy indices $i \leftrightarrow k$ to match the basis.

$\left\{ \begin{array}{l} v^i \Gamma_{ij}^k e_k = \Gamma_{ij}^k v^i e_k \\ \text{rename } k \leftrightarrow i \end{array} \right.$

$$v^i \frac{\partial e_i}{\partial u^j} = \Gamma_{kj}^i v^k e_i$$

put in (A)

$$\frac{\partial v}{\partial u^i} = \frac{\partial v^i}{\partial u^i} e_i + \Gamma_{jk}^i v^k e_i$$

$$\frac{\partial v}{\partial u^i} = \left(\frac{\partial v^i}{\partial u^i} + \Gamma_{jk}^i v^k \right) e_i$$

This show that

$$\boxed{\frac{\partial v}{\partial u^i} = v_{ij}^i e_i} \text{ proved.}$$

where

$$v_{ij}^i = \frac{\partial v^i}{\partial u^i} + \Gamma_{jk}^i v^k$$

Q no 4:- By using Residue theorem, evaluate the integral.

$$\int_0^{2\pi} \frac{\cos 2\theta d\theta}{5 - 4\cos\theta}$$

Solve:-

$$\text{let } z = e^{i\theta}$$

$$dz = e^{i\theta} i d\theta$$

$$dz = z i d\theta$$

$$\boxed{d\theta = \frac{dz}{iz}}$$

$$\rightarrow \cos 2\theta = \frac{1}{2} \left(\frac{z^2 + 1}{z^2} \right)$$

now:-

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$$\int_0^{2\pi} \frac{\cos 2\theta}{5-4\cos\theta} d\theta = \oint_{|z|=1} \frac{1/2 (z^2 + 1/z^2)}{5 - 4 \cdot \frac{1}{2} (z + \frac{1}{z})} \frac{dz}{iz}$$

$$\left\{ \begin{array}{l} \frac{1}{2} \left(\frac{z^4 + 1}{z^2} \right) \Rightarrow \frac{z^4 + 1}{2z^2} \\ 5 - 2 \left(\frac{z^2 + 1}{z} \right) \Rightarrow 5 - \frac{2z^2 + 2}{z} \end{array} \right.$$

$$\frac{5z - 2z^2 - 2}{z}$$

$$\oint_{|z|=1} \frac{1}{2} \left(\frac{z^2 + 1}{z^2} \right) \cdot \frac{1}{5 - 2(z + \frac{1}{z})} \cdot \frac{dz}{iz}$$

multiply and simplify.

$$\frac{1}{2i} \oint_{|z|=1} \frac{z^2 + 1/z^2}{z(5 - 2(z + 1/z))} dz$$

$$\frac{1}{2i} \oint_{|z|=1} \frac{z^4 + 1}{z^3(5 - 2z + \frac{2}{z})} dz$$

Now multiply numerator and denominator by z to eliminate the fraction

$$= \frac{1}{2i} \oint_{|z|=1} \frac{z^5 + z}{z^4(5z - 2z^2 - 2)} dz$$

$$= \frac{1}{2i} \oint_{|z|=1} \frac{z(z^4 + 1)}{z^4(5z - 2z^2 - 2)} dz$$

$$= \frac{1}{2i} \oint_{|z|=1} \frac{z^4 + 1}{z^3(5z - 2z^2 - 2)} dz$$

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$$z=0$$

$$5z - 2z^2 - 2 = 0$$

$$-2z^2 - 5z - 2 = 0$$

$$-2z^2 - 4z - z - 2 = 0$$

$$-2z(z+2) - 1(z+2) = 0$$

$$(-2z-1)(z+2) = 0$$

$$\boxed{z = -\frac{1}{2}}, \boxed{z = -2}$$

$$\text{at } z = -\frac{1}{2}$$

$$R(f, -\frac{1}{2}) = \lim_{z \rightarrow -\frac{1}{2}} \frac{(z+1)}{z} \cdot \frac{z^4+1}{z(-2z+1)(z+2)}$$

$$\lim_{z \rightarrow -\frac{1}{2}} \frac{(z+1)}{z} \cdot \frac{z^4+1}{-2z(z-\frac{1}{2})(-2+2)}$$

$$= \frac{1/16 + 1}{-2(-\frac{1}{2})(-\frac{1}{2}+2)}$$

$$= \frac{-17}{10} \times \frac{2}{5}$$

$$= \frac{-17}{40}$$

$$R(f, -\frac{1}{2}) = \lim_{z \rightarrow 0} \frac{z^4+1}{z(5z-2z^2+2)}$$

$$= \lim_{z \rightarrow 0} \frac{z^4+1}{(-2z+1)(z+2)}$$

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Now

$$\int_0^{2\pi} f(z) dz = 2\pi i (\text{sum of residue})$$

$$= 2\pi i \left(\frac{-17}{40} + \frac{1}{2} \right)$$

$$= \frac{(-17 + 20)\pi}{80} \Rightarrow 3$$

$$= \frac{3\pi}{40}$$

long

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx$$

$$= 2 \int_0^{\infty} \frac{x^4}{1+x^4} dx$$

let $x = t$

$$dx = -\frac{1}{t^2}$$

then

$$\frac{x^2}{1+x^4} dx \Rightarrow \frac{1/t^2}{1+1/t^2} \left(-\frac{1}{t^2} \right) dt$$

$$\Rightarrow \frac{-1}{t^4+1}$$

This symmetry confirms that

$$\int_0^{\infty} \frac{x^2}{1+x^4} dx$$

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$$= \int_0^{\infty} \frac{1}{1+x^4} dx$$

$$2 \int_0^{\infty} \frac{1}{1+x^4} dx$$

Use the standard result.

$$\int_0^{\infty} \frac{1}{1+x^4} dx = \frac{\pi}{2\sqrt{2}}$$

Therefore:

$$\int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx = 2 \cdot \frac{\pi}{2\sqrt{2}}$$

$$= \frac{\pi}{\sqrt{2}} \text{ Ans}$$

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